

Team Name: DraXon

Theme: Geo-Tech

Category: Software

Project Title: Geo-Tagged Waste Reporting and Tracking



Challenge Statement



- Rapid **urbanization** increasing waste generation
- Waste reporting systems are **manual**
- Complaints are **text-based** and **lack visual evidence**
- There is **no priority system** for hazardous waste cases
- Delayed response lead to **environmental & public health risks**



Real-World Scenario



Delhi faces a deluge of waste on its roads

1. With its overloaded landfills and lack of coordination between various stakeholders, Delhi continues to have an inefficient waste management system.
2. The city generates over 11,000 tons of waste daily. However, it faces significant challenges due to poor waste segregation.



Real World Scenario



Waste Dumping in Tamil Nadu

1. Tamil Nadu generates about 14,200-15,200 tonnes of municipal solid waste per day.
2. The state processes around 66% of this waste.
3. The remaining ~34% (~5,000 tonnes per day) ends up in landfills or unmanaged places which includes roadsides, open lands, drains, etc.



Proposed Solution



- Users report waste by **uploading photos**
- The system automatically captures the exact **geo-location**
- **AI verifies the image**, identifies waste type category
- Multiple reports from the same geo-tagged location **increases priority**
- Authorities view real-time **waste hotspots**
- Cleaning staff receive assigned tasks with location
- After cleaning, staffs upload an **after-clean image** and **awards credit points**



Technical Description



- Frontend: **React Native**
- Backend: **Node.js**
- Database Management: **MongoDB**
- Waste vs Non-Waste image detection: **MobileNetV2**
- Waste type classification: **YOLOv8**
- Before–after cleanliness verification: **Siamese Neural Network**
- Location tracking: Google Maps **API**



System Design



Fig.1.1 System Workflow



Fig.1.2 ML Waste Prediction



Impact & Benefits



- **Speed:** Faster reporting, verification, and cleanup
- **Cost:** Reduced manual work and unnecessary field visits
- **Simplicity:** Easy-to-use system for citizens and cleaners
- **Accessibility:** Can be used by anyone with a smartphone
- **Accuracy:** AI verification and geo-tagging improves accuracy
- **Transparency:** Clear confirmation through after-clean photos



Feasibility & Viability



- AI image recognition validates waste reports
- Automatic geo-tagging ensures precise location tracking
- Users earn credit points for submitting waste reports, encouraging active participation
- Cleaners receive credit points only successful cleaning
- Credit points can be redeemed as cash incentives
- Requires only a smartphone with internet access



Competitor Analysis



- **Swachhata App (India)**

- ✓ Allows citizens to upload waste complaints with image and geo-location
- ✗ Does not use AI verification and lacks accountability for timely cleaning.

- **Our AI-Driven Approach**

- ✓ Uses AI to verify waste and assign priority
- ✓ Multiple reports from the same geo-tagged location **increases priority**
- ✓ Authorities view real-time **waste hotspots**
- ✓ AI verifies cleanliness and **awards credit points to cleaners**



Limitations of Existing System



- Manual Reporting Process
- No AI Verification
- Lack of Priority Handling
- Delayed Response Time
- No Real-Time Tracking

